

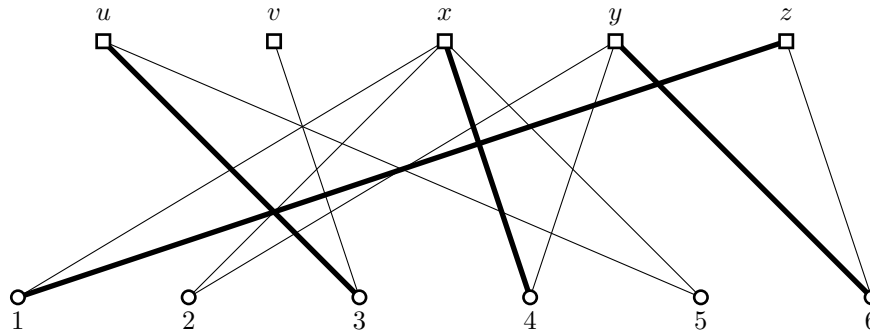
COMP1870 Theoretical Foundations of Computer Science I

School of Computer Science, University of Leeds

Worksheet - Week 1.1 submission template

Questions

5 Let G be the graph shown below.



Let X be the set of vertices marked by circles, and let Y be the set of vertices marked by squares. Let M be the matching consisting of thicker edges.

Apply Algorithm 1 from the lecture notes to this input to find an augmenting path. Start by setting L to be the list $[2, 5]$ and marking every vertex as unvisited except for the vertices in L . Every time we go through the **while** loop between lines 5 and 31, we change the list L . Write down L each time you reach the end of this loop. When you process the unvisited vertices in line 9, process them in alphabetical or numerical order. What is the augmenting path produced as output by the algorithm?

Write your solution in this box:

10 Use the logical equivalences in Section 2.1 of the lecture notes to show that

$$\neg((P_1 \wedge P_2) \vee (P_2 \Rightarrow P_3)) \text{ is equivalent to } (((\neg P_1) \vee (\neg P_2)) \wedge P_2) \wedge (\neg P_3)$$

At each step, state which equivalence you are using.

Write your solution in this box: